National University of Computer and Emerging Sciences, Lahore Campus

Course:	**COAL**	**Course Code:**	**EE2003**
Program:	**BS(CS)**	**Semester:**	**Fall 2024**
Duration:	**1 Hour**	**Total Marks:**	**40**
Paper Date:	**24-09-2024**	**Page(s):**	**4**
Section:	**All**	**Roll No.**	
Exam:	**Midterm-I (Retake)**	**Your Section:**	

Instruction/Notes: **This is an open book exam. Sharing book or calculators is NOT ALLOWED. PAPERS/NOTES ATTACHED WITH BOOK ARE NOT ALLOWED.** Rough sheets can be used but will not be collected and checked. In case of any ambiguity, make reasonable assumptions. Questions during exams are not allowed.

Question 1 [CLO 1] [10 Marks]: Short questions

i) **[2 Mark]** Give two different segment-offset pairs for a memory location whose physical address is 0x1256A _____ , _____

ii) **[2 Mark]** For each of the instructions given below, **identify whether the instruction is valid or invalid:**

Instruction	Valid/Invalid	Instruction	Valid/Invalid
mov [bx+si],10		mov byte [mem1],al	
mov [bp-0x0002],2		mov [Bp-si],6	

iii) **[2 Mark]** For following write the required code, assume that least significant bit is bit number 0:

a) Using one instruction, set bits 5 and 6: MOV AL, 0xFF _____	**b)** Using one instruction, divide al by 4 (DIV not allowed): MOV AL, 0xAA _____

iv) **[4 Mark]** In the following code, tell whether the jump will be taken or not. Show each and every step to get marks.

[org 0x100]
mov ax, 0x8B23
mov bx, 0x4321
sub ax, 0x1234
cmp ax, bx
je exit
exit:
mov ax, 0x4c00
int 21h
	Working:

Question 2 [CLO 2] [10 Marks]:

i) **[6 Mark]** Given the following jump statements and their opcodes, identify the type of jump (near or short), the offset (value of IP) to which the jump will take place and the physical address to which the jump will take place.

CS: 0x1D3F, DS: 0x1D3D, SS: 0x1D3E

Show your complete working to get credit.

Offset of Opcode	Opcode	Type of Jump	Offset	Physical Address
02A9	77F1			
0B9E	E98811			

Working:

ii) **[4 Mark]** What is size of the following file in bytes.

```
1  [org 0x0100]
2  00000000  A1[1700]      mov ax, [num1]
3  00000003  8B1E[1900]    mov bx, [num2]
4  00000007  01D8          add ax, bx
5  00000009  8B1E[1B00]    mov bx, [num3]
6  0000000D  01D8          add ax, bx
7  0000000F  A3[1D00]      mov [num4], ax
8
9  00000012  B8004C        mov ax, 0x4C00
10 00000015  CD21          int 0x21
```

Working:

Question 3 [CLO 3] [20 Marks]: Write assembly language program to find Mode of an array i.e. the most frequent number in the array. Safely assume that input array will be sorted in ascending order and it will always have exactly one mode. Sample run is shown below:

Sample Run:

Example 1	Arr: 1,1,2,2,2,2,4,4,4,4,4	Mode = 4
Example 2	Arr: 1,1,2,2,2,2,3,4,4,4,5	Mode = 2

Solution:

```
[org 0x0100]
jmp start
arr: db 1,1,2,2,2,2,4,4,4,4,4
count: dw 11 ; There are total 11 elements in array
; add extra variables here, if required

start:
```